

Multi-Dimensional Array

The 2D arrays, 3D arrays, etc are called Multi-dimensional Arrays.

Two Dimensional Array

If an array contains more than 1 row and 1 column that is known as Two Dimensional Array. It is also known as array of arrays.

Ex:-

```
a = array([ [10, 20, 30, 40],  
            [50, 60, 70, 80] ])
```

Similarly we can create 3D array.

Ex:-

```
a = array([ [ [2, 5, 8], [6, 4, 3] ],  
            [ [3, 7, 9], [5, 4, 9] ] ])
```

Way of creating Multi-D Array

- array () Function
- zeros () Function
- ones () Function
- reshape () Function

array () Function

array () Function of numpy is used to create a multi dimensional array.

Syntax:-

```
numpy.array(object, dtype=None, copy=True, order='K', subok=False, ndmin=0)
```

<https://www.numpy.org/devdocs/reference/generated/numpy.array.html>

Creating 2D Array using array () Function

Syntax:-

```
import numpy
```

```
array_name = numpy.array([ [elements], [elements] ])
```



array Function

Ex:-

```
import numpy
```

```
a = numpy.array([ [10, 20, 30, 40], [50, 60, 70, 80] ])
```

```
a = numpy.array([ [10, 20, 30, 40],  
                  [50, 60, 70, 80] ])
```

```
a = numpy.array([ ['Rahul', 'Sonam', 'Raj'],  
                  ['Dell', 'Asus', 'Lenovo'] ], dtype=str)
```

Creating and Initializing 2D Array

Syntax:-

```
from numpy import *
```

```
array_name = array([ [elements], [elements] ])
```

Ex:-

```
from numpy import *
```

```
a = array([ [10, 20, 30, 40],  
            [50, 60, 70, 80] ])
```

index

An index represents the position number of an array's element.

```
from numpy import *
```

```
a = array([ [10, 20, 30, 40],  
           [50, 60, 70, 80] ])
```

Python interpreter allocates 8 blocks of memory and stores the elements



Index always starts with 0

	0	1	2	3
0	10	20	30	40
1	50	60	70	80

10	20	30	40	50	60	70	80
[0][0]	[0][1]	[0][2]	[0][3]	[1][0]	[1][1]	[1][2]	[1][3]

Accessing 2D Array Elements

```
from numpy import *
```

```
a = array([ [10, 20, 30, 40],  
           [50, 60, 70, 80] ])
```

```
print(a[0][0])
```

```
print(a[0][1])
```

```
print(a[0][2])
```

```
print(a[0][3])
```

```
print(a[1][0])
```

```
print(a[1][1])
```

```
print(a[1][2])
```

```
print(a[1][3])
```

10	20	30	40	50	60	70	80
a[0][0]	a[0][1]	a[0][2]	a[0][3]	a[1][0]	a[1][1]	a[1][2]	a[1][3]

Modifying 2D Array Elements

```
from numpy import *
```

```
a = array([ [10, 20, 30, 40],  
           [50, 60, 70, 80] ])
```

```
a[1][2] = 100
```

```
print(a[1][2])
```

10	20	30	40	50	60	70	80
a[0][0]	a[0][1]	a[0][2]	a[0][3]	a[1][0]	a[1][1]	a[1][2]	a[1][3]

Accessing 2D Array using for loop

```
from numpy import *
```

```
a = array([ [10, 20, 30, 40],  
           [50, 60, 70, 80] ])
```

Without index

```
for r in a:  
    for c in r:  
        print(c)  
    print( )
```

Outer for loop

Inner for loop

Inner for loop

With index

```
n = len(a)  
for i in range(n):  
    for j in range(len(a[i])):  
        print(a[i][j])  
    print( )
```

Outer for loop

The outer for loop represents the rows and the inner for loop represents the columns in each row.

Accessing 2D Array using while loop

```
from numpy import *  
a = array([ [10, 20, 30, 40],  
            [50, 60, 70, 80] ])   
n = len(a)  
i = 0  
while i < n :  
    j = 0  
    while j < len(a[i]):  
        print(a[i][j])  
        j+=1  
    i+=1
```